Test-beam topology and response-model studies of scintillator range staves for HIBEAM/NNBAR

HIBEAM/NNBAR CCB test-beam working group
Working publication package

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Working publication – not submission-authorising. This PDF is the canonical chapterised build surface for the paper, but the Cycle-3 adversarial audit on `main@32fd72b675aa4cba0da3bdee52561755be960ada` invalidates promotion of the current ΔE–E production numbers, the July optical calibration grid, and the 8.9% deposited-energy-resolution headline. Those quantities are kept only in clearly marked publication-hold text until their physics/provenance issues close. The authoritative status is tracked in the repository claim ledger and publication audit files.

Contents

Abstract

The HIBEAM/NNBAR programme is developing an annihilation detector for neutron-conversion searches, including free neutron–antineutron oscillation. One proposed detector element is a range stack of plastic-scintillator staves intended to sample the penetration depth and longitudinal energy-loss pattern of low-energy charged particles. This paper studies the CCB test-beam range-stack prototype using beam data and Geant4-based detector simulations.

The analysed B-stack readout uses channels labelled B2, B4, B6 and B8, while the corresponding detector model contains eight physical longitudinal layers. Existing source-bound selected-data products indicate different longitudinal occupancy patterns for the declared Sample-I and Sample-II run families, with one population concentrated closer to the stack entrance and the other extending more frequently to downstream readout positions. The depth profile has been regenerated from the full pre-threshold 8-channel-by-16-sample event product (33 runs, 1.1M events, run-block bootstrap uncertainty): Sample I concentrates at the entrance (87.4% of normalized amplitude in B2) while Sample II extends further downstream (6.1% in B8, $\approx 7.6\times$ Sample I). The quantitative result remains gated — the profile shape is strongly threshold-dependent and the threshold/baseline contract is not yet closed — and primary run-log and hardware-trigger records are incomplete, so the difference is interpreted as a run-family observation rather than a measured trigger efficiency.

The physically relevant amplitude telescope observable is $\Delta E = A(B2)$ and $E = A(B4) + A(B6) + A(B8)$. Sparse longitudinal readout can hide the layer containing the largest energy loss and therefore distort a stopping/range proxy. The current numerical ΔE–E data/MC bundle is under publication hold because the beam event construction is threshold-censored upstream and the MC producer does not preserve an unambiguous physical-layer namespace or validated event-measure contract.

The located beam waveform product contains eight channels with 16 samples per channel. A component-safe B4–B6 analysis of the authorising Sample II population (10,776 events) gives a pair-residual central width of 8.7 ns (bootstrap interval 8.3–9.3 ns), superseding an earlier 38 ns direct-mode diagnostic. This is reported only as a waveform-format/estimator-level pair residual: component identity, timing reference, run dependence and covariance do not support conversion to an intrinsic single-stave resolution.

A detailed single-stave optical model provides a framework for separating quenching, scintillation, wavelength shifting, optical transport, SiPM response and electronics. The historical July optical grid and its derived held-out energy-reconstruction result are superseded model diagnostics because the campaign predates source-level optical repairs and because its target variable was Birks-visible energy rather than unquenched deposited energy. The calibration grid has since been regenerated on the repaired source with explicit E_{raw} and E_{vis} semantics, and the held-out reconstruction has been rebuilt on it with propagated uncertainty; both remain MC_MODEL_DEPENDENT diagnostics, quarantined from validated promotion until the optical/SiPM nuisance envelope is propagated, held-out coverage spans more operating points and impact-position transfer is tested beyond the central line. Absolute beam-data-calibrated light yield and detector energy resolution therefore remain withheld.

1 Introduction

The HIBEAM/NNBAR programme at the European Spallation Source is developing the experimental techniques required for searches for baryon-number-violating neutron-conversion processes, including free neutron–antineutron oscillation [1, 3]. In a neutron–antineutron search, an antineutron would annihilate in a thin target and produce a low-energy hadronic final state whose charged particles, pions and secondary nuclear interactions must be reconstructed with high efficiency while backgrounds are rejected.

These requirements differ from those of a conventional collider spectrometer. The neutron flight region must remain magnetically quiet, restricting the use of magnetic momentum analysis close to the interaction region, while many charged annihilation products are sufficiently slow that range and specific-ionisation information are directly useful. The HIBEAM/NNBAR calorimeter concept therefore includes multiple layers of plastic scintillator ahead of the electromagnetic calorimeter, so that the longitudinal pattern of deposited energy and the depth reached by a charged particle can supplement calorimetric and timing information [2]. A segmented scintillator range stack also provides a controlled system in which the consequences of sparse readout, thresholding and finite waveform sampling can be studied explicitly.

The CCB test-beam data provide an experimental test of this range-stack concept with charged-particle events that can also be represented in Geant4. The present study focuses on five connected questions:

(i) what longitudinal-response differences are observed between the declared beam-data run families;

(ii) how much range and energy-loss information is lost when the analysed readout samples only alternating positions in the B stack;

(iii) what timing observable is supported by the source-bound waveform format and estimator;

(iv) which causal response stages must be represented in a one-fibre, one-end scintillator/WLS/SiPM model; and

(v) how raw deposited energy and Birks-visible energy should be kept distinct when defining and validating an energy-reconstruction estimator.

The analysis intentionally separates direct beam-data observables from truth-level simulation and from model-dependent optical/electronics predictions. This distinction is essential for the present detector state: a result that is reproducible within a specified Monte-Carlo model is evidence about that model, but it becomes a detector-performance measurement only after the relevant hardware, calibration, data-lineage and held-out validation conditions are established. The evidence and provenance rules used to enforce this separation are described in the reproducibility section rather than being treated as additional detector observables.

2 CCB test-beam configuration

Hardware facts, simulation-configuration values and unresolved legacy narratives are separated in the publication hardware bill of materials. Each quantity should carry one of four statuses: MEASURED, DESIGN_SPEC, SIM_CONFIG, or UNKNOWN_EXTERNAL. Simulation agreement is not hardware provenance.

2.1 Beam, target and two-arm layout

The reviewed CCB simulation configuration uses a 190 MeV incident proton beam and a deuterated target. The model contains the elastic channel

$$p + d \to p + d,$$

and transports the outgoing charged particles toward two scintillator arms. Historical source configuration uses a nominal geometry-distance parameter of 109 cm, with the B stack near $-38°$

and the A stack near +71.5°, and contains eight B bars and four A bars. These values are publication-safe as simulation/configuration quantities; they are not survey-grade metrology until matched to primary collaboration records. An external collaboration preprint independently reports the in-beam CCB configuration — a 190 MeV proton beam on a CD_2 target, a TPC arm at 71° to the beam and a second scintillator arm at 38°, with two-arm coincidences selecting candidate elastic-scattering events [9]; the reported arm-angle magnitudes are consistent with the two-arm configuration above.

The compact detector simulation contains the target, scintillator stack volumes, trigger-related volumes and ProtoTPC components. Source review shows that it is useful for first-order acceptance and truth-energy-deposit studies but does not yet represent every passive material relevant to precision stopping predictions. Wrapping, supports, photosensor boards, cabling, dead layers and survey uncertainties must therefore be treated explicitly when the paper makes material-budget claims.

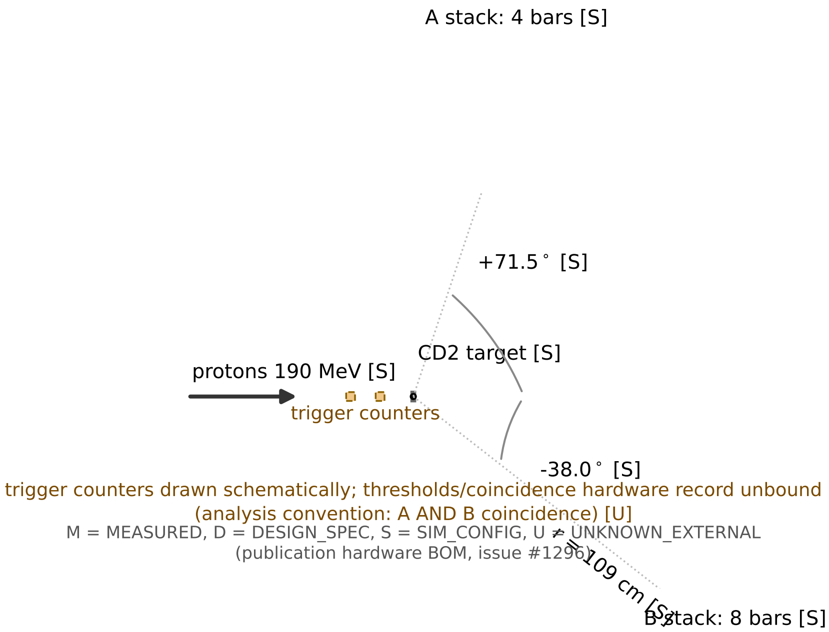

Figure 1: CCB two-arm geometry schematic. Beam energy, target, arm angles and bar counts are annotated on-figure with per-quantity evidence status from the hardware truth surface; they are simulation/configuration values, not survey-grade metrology.

2.2 B-stack readout and run families

The analysed data use four B-stack channels labelled B2, B4, B6 and B8. The simulation contains eight physical B layers, so the data represent alternating longitudinal samples. The exact offset between detector labels and Geant4 copy/layer IDs remains a publication blocker: historical issue #869 argued for layers 1/3/5/7 while later paper/BOM material uses 0/2/4/6. Only the every-other-layer structure is currently robust. A final data-matched MC figure must import a single source-bound detector map rather than hard-code either convention.

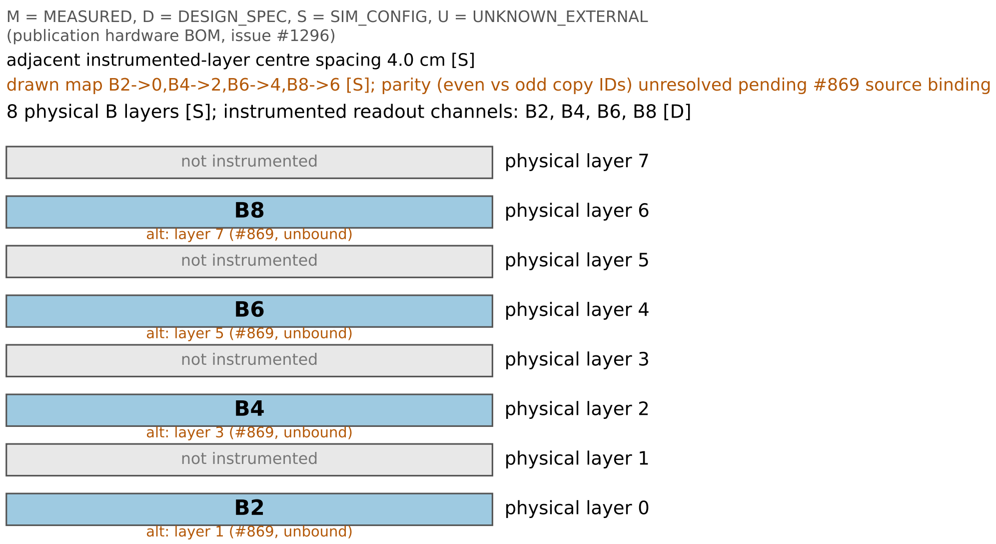

Figure 2: B-stack channel-to-layer correspondence. Channels B2/B4/B6/B8 sample every other physical Geant4 layer; the absolute offset between detector labels and copy IDs (the even-versus-odd parity of issue #869) is annotated as unresolved rather than silently chosen.

The current analysis configuration separates Sample-I and Sample-II run families and also separates calibration and analysis roles. However, the exact hardware trigger thresholds, coincidence logic, prescales and run-purpose record remain partly external. The safest beam-data language is therefore *Sample-I/Sample-II run-family comparison*. The MC selection based on first-stack-layer charged hits is retained only as `MC_TRIGGER_PROXY` until the hardware response is modelled or demonstrated equivalent.

> **PUBLICATION HOLD.** Do not use the words “trigger efficiency”, “hardware-trigger reproduction”, or quantitative trigger-induced species fractions until issues #962 and #1045 close on the same run/hardware truth surface (#1296 closed scope-bound 2026-08-14; its hardware-UNKNOWN rows remain recorded in the BOM).

3 Scintillator stave and readout

3.1 Stave geometry

The current design specification follows the later collaboration clarification associated with issue #796. Each B stave is described as an extruded-polystyrene scintillator approximately 50 cm long, 5.18 cm wide and 2.0 cm thick along the nominal particle path. Two longitudinal holes of 2.0 mm diameter are separated by 2.0 cm centre-to-centre and accept 1.8 mm Kuraray Y-11 wavelength-shifting fibres. The exterior includes a reflective coating. Older repository prose describing approximately one-metre BC-408 bars is treated as superseded/unknown external history rather than an alternative current specification.

The stave design supports two longitudinal fibres and, in the full concept, readout from both fibre ends, corresponding to four possible optical measurements. The later collaboration clarification states that the CCB beam-test configuration used one fibre at one end; in the present evidence hierarchy this remains a DESIGN_SPEC statement until the installed hardware/BOM record closes under issue #1296. If confirmed for the analysed hardware, one-ended readout makes the response dependent on longitudinal hit position and removes the charge/time asymmetry that a two-ended system could use to constrain position.

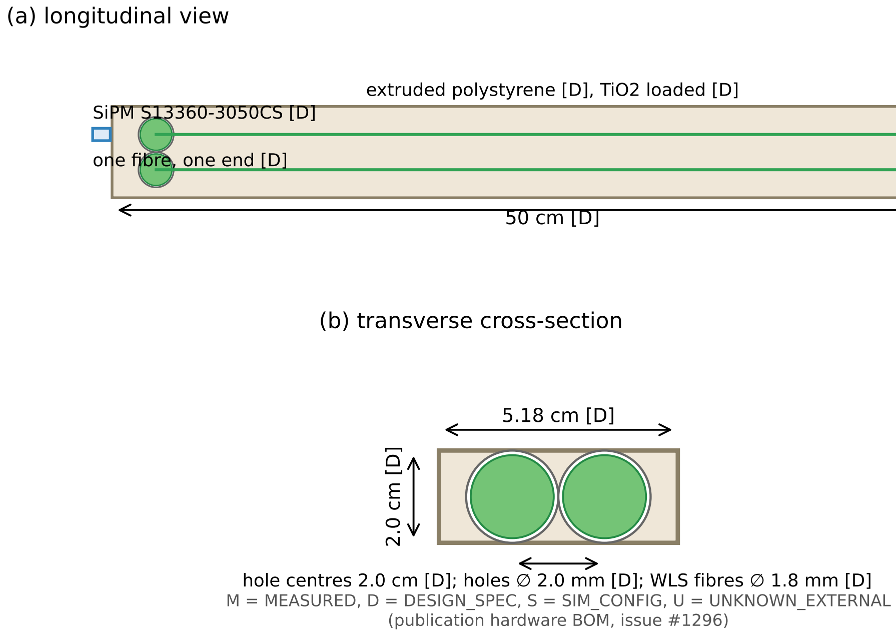

Figure 3: B-stack stave geometry (design specification, issue #796). Dimensions, hole and fibre parameters and the one-fibre one-end readout configuration are annotated on-figure with evidence status; of the four possible fibre-end measurements only one was instrumented, and the reflective outer coating enters the optical model as a distinct surface.

3.2 SiPM and electronics boundary

The detector model uses a Hamamatsu S13360-3050CS MPPC. Manufacturer data specify a 3 mm×3 mm photosensitive area, 50 μm pixel pitch and 3600 pixels, with spectral sensitivity covering the Y-11 green-emission region [8]. These public specifications do not by themselves determine the CCB operating response. PDE, overvoltage, temperature, optical coupling, illumination footprint, microcell recovery, crosstalk, afterpulsing, gain and the analogue impulse response are separate response objects.

The publication model therefore keeps sensor photons, primary avalanches, correlated avalanches, charge and ADC waveform observables distinct. A scalar efficiency is not allowed to absorb unresolved physics from multiple stages.

3.3 Waveform products

Two waveform schemas appear in the project history. Historical reconstruction configurations use 18 samples per channel, whereas the located LUNARC beam product contains eight channels with 16 samples per channel (128 waveform words per event). The provenance analysis under issue #993 classifies these as distinct schemas rather than a proven reversible 16-to-18 sample transformation. Consequently, historical sub-nanosecond timing studies based on the 18-sample product cannot be promoted to the source-bound 8-by-16 beam sample.

Every publication amplitude/timing figure must record waveform schema, raw-file hashes, producer revision, channel map and pulse-polarity contract. The publication source package intentionally does not hide these provenance requirements behind a generic label such as "raw data".

4 Simulation programmes

4.1 CCB event simulation

The CCB event simulation provides particle identities, trajectories and energy deposits across the two-arm detector. Historical production used Geant4 electromagnetic and hadronic/ion transport suitable as a starting point for proton/deuteron studies in this energy domain [5, 6]. The simulation is used for truth-level stopping and for testing how detector segmentation changes observables.

The elastic p–d angular source table used by the current source-model work is bound to the 190-MeV centre-of-mass differential cross sections reported in Table VI of Ermisch et al. [4]. The deterministic inverse-CDF implementation defect tracked by issue #1178 has been repaired: the current source explicitly defines its interpolation and measured-support policy and can generate the nominal central-value law with unit event weights for direct-sampled campaigns. This bounded source-sampler closure must not be confused with publication-MC closure. Source-model uncertainty, surviving support/model alternatives, exact production provenance and downstream detector-response dependencies remain separate requirements.

Legacy samples can contain non-unit `PrimaryWeight` values from a generator source-measure implementation whose frame/Jacobian semantics were later found to be defective. The exact paper MC file `output_krakow_1M.root` is byte-hashed but is not yet bound to a complete production receipt containing generator revision, cross-section table/mode, seed, physics list, build/toolchain and all geometry/material configuration. That legacy file therefore remains a historical provenance-gated diagnostic rather than the nominal publication MC. A regenerated 100,000-event campaign closing issue #1311 now carries a complete production receipt (generator source-pair and cross-section-table digests with the direct-CDF sampling mode, executable and runtime-stack identity, serial single-worker configuration) in `geant4/manifests/cmc_100k_regenerated_20260814.json`; engine seeding is recorded honestly as default per-run seeding.

4.2 Single-stave optical simulation

The single-stave model transports the charged particle through a detailed scintillator/fibre geometry and follows optical photons through scintillation, wavelength shifting, attenuation and sensor boundaries. A separate SiPM/electronics layer converts sensor arrivals into avalanche/charge/waveform quantities.

The model is scientifically most useful when represented as a causal graph:

$$E_{\mathrm{raw}} \to E_{\mathrm{vis}} \to N_{\gamma,\mathrm{scint}} \to N_{\gamma,\mathrm{WLS\,abs}} \to N_{\gamma,\mathrm{WLS\,emit}} \to N_{\gamma,\mathrm{sensor}} \to N_{\mathrm{primary\,aval}} \to Q \to A_{\mathrm{ADC}}.$$

Here E_{raw} is unquenched deposited energy and E_{vis} is the Birks-visible energy. These two variables must never share the same bare E_{dep} label.

The current single-stave source explicitly separates the fibre inner and outer cladding materials and records the WLS secondary-multiplicity parameter in the optical configuration. These implementation repairs post-date the July optical calibration grid used by the historical response studies. Therefore the old grid is a superseded model diagnostic; it has been regenerated from the current source (#1303, complete 2026-08) with explicit E_{raw} and E_{vis} semantics, and promotion of absolute optical-response or energy-reconstruction numbers remains gated on propagating the optical/SiPM nuisance envelope and on broader held-out coverage.

4.3 Simulation evidence classes

Paper figures distinguish `TRUTH_LEVEL_MC`, `DIGITISED_MC`, `MODEL_DEPENDENT_OPTICAL_MC`, and `CALIBRATED_DETECTOR_MC`. The last category is intentionally empty until response nuisances and held-out data closure satisfy the publication gate.

5 Data taking and event selections

5.1 Run families

Current configuration files define Sample-I and Sample-II analysis populations and separate calibration populations. These configuration roles are useful for reproducible processing but are not a substitute for the primary hardware run log. The run-role discrepancy discussed under issue #962 means that calibration/analysis assignment must be bound to the exact fitted object before timing or response calibration can be described as held out.

The historical S00 selected-pulse product contains 640,737 B-stack pulses. That count is deterministic for the fixed historical producer/input set, but the canonical claim ledger keeps this claim GATED (promotion blocked) because the raw waveform width, raw-to-sorted lineage and channel-polarity/hardware contracts are not all independently closed. Selected-pulse counts are not detector efficiencies.

5.2 Longitudinal topology in the existing selected products

Existing source-bound selected-data studies show different B-stack occupancy patterns for the declared Sample-I and Sample-II populations: one population is more concentrated near the entrance while the other contains a larger downstream fraction. This is the central beam-data range-stack result. The pre-threshold 8-channel-by-16-sample event product has been regenerated and analysed (#1318): 33 runs and 1,096,728 events (Sample I 798,694; Sample II 298,034) with frozen amplitude/polarity definitions and a run-block bootstrap (1,000 replicates, seed 1318). The normalized amplitude profiles concentrate Sample I at the entrance (B2 87.4%, B8 0.8%) while Sample II extends further downstream (B2 72.7%, B8 6.1%; deepest-stave share $\approx 7.6\times$ Sample I). A threshold scan at 500/750/1000 ADC preserves the direction of the difference while strongly compressing its magnitude (B8 ratio ≈ 1.3–$1.4\times$), so the quantitative profile remains GATED in the claim ledger (CL-1318-001; `allowed_status_validated=NO`) until a threshold/baseline contract closes.

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/final/data_depth_profile.pdf`

Regenerated product delivered (#1318): run-block bootstrap with 1,000 replicates and a 0–1000 ADC threshold scan; the claim remains GATED pending threshold-contract closure, so the figure is not rendered in the publication build. Artifact bundle: `reports/studies/paper_1318_depth_profile/`.

Figure 4: Beam-data B-stack depth profile for Sample-I and Sample-II from the pre-threshold 8-by-16 event product (CL-1318-001, GATED).

Evidence boundary. Until the final raw-event rerun and hardware run/trigger record close, the paper should call the existing difference a run-family-dependent longitudinal-topology indication. Interpreting it uniquely as a trigger effect would confound trigger changes with possible run-period beam/electronics changes, and calling the deepest active readout a stopping layer would additionally assume a range convention that the sparse readout does not itself establish.

5.3 Monte Carlo population comparison

The corresponding MC population is currently formed with `MC_TRIGGER_PROXY`. It can show that the transport model contains physically plausible stopping/penetrating populations, but it cannot authorise a hardware-trigger efficiency or a precise proton/deuteron mixture until event-level species counting, trigger response, source measure and production provenance all close.

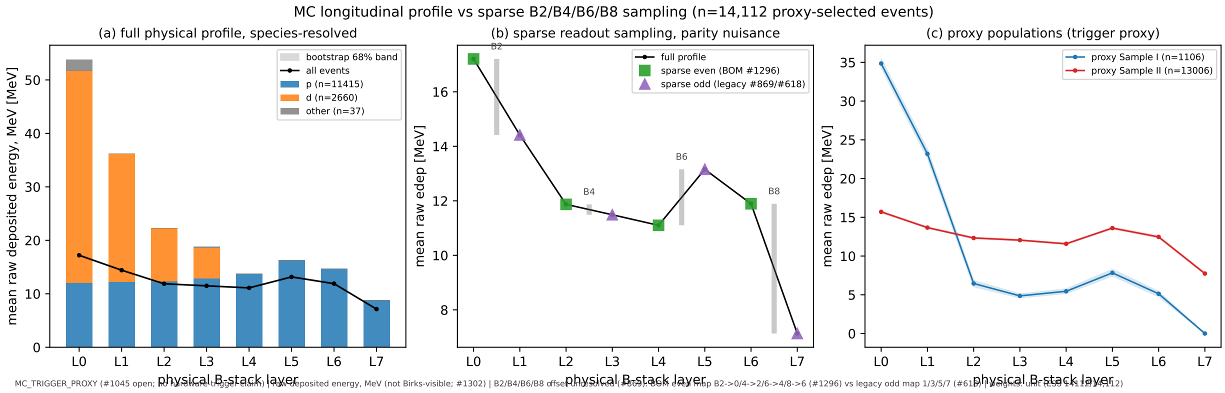

Figure 5: MC longitudinal deposited-energy profile of the proxy-selected population ($n = 14{,}112$, weights verified unit, ESS $= 14{,}112$): (a) per-layer mean raw deposited energy, species-resolved, with event-bootstrap 16–84% bands on the total; (b) the sparse B2/B4/B6/B8 sampling of the same profile under both parity hypotheses of issue #869, drawn as a nuisance envelope because the channel-to-layer offset is unresolved (neither the BOM even map #1296 nor the legacy odd map carried by the #618 readout columns is canonical); (c) the proxy Sample I/II populations. Energy is raw deposited MeV, not Birks-visible (#1302).

6 Inter-stave timing

6.1 Observable and covariance

For staves i and j, define

$$\Delta t_{ij} = t_j - t_i - t_{\mathrm{TOF},ij}.$$

The pair width does not uniquely determine single-stave resolutions. In general,

$$\mathrm{Var}(\Delta t_{ij}) = \sigma_i^2 + \sigma_j^2 - 2\,\mathrm{Cov}(t_i, t_j).$$

Common trigger phase, sampling phase, reconstruction choices and event conditions may produce covariance. The paper therefore does not divide a pair width by $\sqrt{2}$ without an independently justified model.

6.2 Pulse-component definition

Leading-edge timing is amplitude-dependent. Constant-fraction timing also becomes ambiguous for multi-component waveforms if the threshold is defined from the global maximum: changing the fraction can switch the timed physical component. The production method must define the selected pulse component before optimising any timing fraction. Same-sample minimum widths across a fraction scan are exploratory model-selection diagnostics, not performance measurements.

6.3 Current 8-by-16 data diagnostic

On the located 8-by-16 raw waveform product, the current direct B4–B6 CFD reconstruction returns 5,207 events with valid times and an approximately 38 ns central-68% residual width (format-/reconstruction-limited; see the evidence box below) after the applied TOF subtraction. Peak positions are multi-modal and frequently lie near waveform boundaries. The component-safe rerun has now been executed (#1335): on the complete authorising Sample II population (runs 58–63 and 65; 10,776 events), the component-safe constant-fraction mode (20% fraction, first-local-peak component selection per #1059) with the 0.312 ns B4–B6 time-of-flight correction yields a pair-residual central-68% width of 8.748 ns (1,000-replicate bootstrap interval 8.295–9.270 ns) and a median of −44.254 ns, superseding the 5,207-event direct-mode diagnostic above. Publication promotion remains conditional: the synthetic two-pulse component-recovery validation fails (0.0% fraction correct), so multi-component identity is not validated, and timing/TOF definitions remain analysis contracts rather than source-bound hardware quantities.

> **Evidence boundary.** The pair-residual width (8.748 ns in the component-safe rerun, superseding the earlier 38 ns direct-mode diagnostic) is reported as a **format-/reconstruction-limited pair residual**. The nominal 10 ns sampling interval is itself comparable to the rerun width, and window phase, pulse-component ambiguity (the synthetic two-pulse validation fails), boundary truncation and timing-reference/TOF assumptions remain entangled. No intrinsic stave timing resolution is extracted, and the residual width is not divided by $\sqrt{2}$. The timing-residual calculation is governed by `scripts/issue_1320_timing_residual.py` with a MANIFEST entry tracking source provenance (#1343); the authorising artifact is committed under `reports/issue_1320_timing/` and bound as a gated figure below.

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/final/timing_b4_b6_residual.pdf`

Component-safe rerun delivered (#1335, #1320): central-68% width 8.748 ns (bootstrap 8.295–9.270 ns), 10,776 Sample II events, TOF-corrected; synthetic two-pulse component recovery failed, so the claim stays GATED and the figure is not rendered in the publication build. Artifact: `reports/issue_1320_timing/`.

Figure 6: B4–B6 pair timing residual from the source-bound 8-by-16 waveform product (Sample II, component-safe CFD20, first-local-peak mode). NOT DETECTOR RESOLUTION.

Historical sub-nanosecond values are retained only as provenance history or simulation/method-closure diagnostics and must not appear in the abstract/conclusions as beam-data performance.

7 Amplitude ΔE–E and the segmentation limit

7.1 Observable definition

The supervisor-defined beam-data analogue of a range-telescope observable is

$$\Delta E_{\text{data}} = A(B2), \qquad E_{\text{data}} = A(B4) + A(B6) + A(B8),$$

where A is a baseline-subtracted ADC amplitude. These axes are response proxies, not calibrated energies.

A data-matched MC view should use the same four physical readout positions,

$$\Delta E_{\mathrm{MC},4} = E_{\mathrm{dep}}(B2), \qquad E_{\mathrm{MC},4} = E_{\mathrm{dep}}(B4) + E_{\mathrm{dep}}(B6) + E_{\mathrm{dep}}(B8),$$

while a full-segmentation truth view must sum each unique downstream physical layer exactly once. A B2-versus-B4 plot is a two-channel response diagnostic and must not be called ΔE–E.

7.2 Sparse-segmentation mechanism

Alternating readout positions can miss the layer in which the stopping-power rise becomes largest. A change in entry energy, angle or passive material can move the Bragg-rise region between sampled and unsampled layers and thereby alter the apparent sparse ΔE–E topology. The publication test is therefore a controlled comparison of full physical layers and the exact data-matched readout mask, with readout phase treated as a nuisance until the hardware map is source-bound.

7.3 Cycle-3 status of the current production bundle

> **PUBLICATION HOLD.** The numerical bundle under `reports/paper_956_deltaE_E_20260814T090700Z/` remains **not** publication-authorising. The producer was repaired in #1336: (i) beam-data input now uses the authorising 8-by-16 raw schema; (ii) threshold-censoring order is corrected; (iii) physical-layer and readout namespaces are kept distinct. Corrected-run evidence is recorded in claims CL-030..CL-033 (#1342). Gate status as of 2026-08-14: (iv) the channel-to-layer map remains a documented simulation contract, not a hardware-source-bound drawing (BOM row `B_channel_to_G4_layer_map`; #1296 closed scope-bound); (v) the saturation and transfer threshold remains a software contract without hardware source-binding (the ADR records BLOCKED as a preserved blocker for #1014/#1073 — no hardware identity is invented); (vi) MC production provenance is closed: the regenerated 100,000-event campaign carries a complete production receipt (#1311 closed via #1354; `geant4/manifests/cmc_100k_regenerated_20260814.json`). The quarantine on the OLD bundle stays; no promotion to VALIDATED is authorized until all gates pass and a final figure package is delivered.

The quarantined plots are kept under `publication/figures/gated/` to preserve provenance without allowing accidental use in the paper.

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/gated/fig07_data_deltaE_E_per_sample.png`

GATED by Cycle-3 audit; do not cite as final result.

Figure 7: Quarantined current beam-data amplitude ΔE–E plot.

The final Figure 7/8 production must start from an event-level table that contains all four reconstructed channel amplitudes before analysis-threshold censoring, then apply declared threshold scans; import the source-bound readout map; keep physical-layer and readout namespaces distinct; and use a production MC whose source measure and run provenance are authorising.

7.4 Species-resolved topology and penetration deliverables (#618)

The supervisor-requested species and penetration set is delivered from the corrected event tables of the regenerated-campaign producer. Species-resolved ΔE–E panels (proton truth, deuteron truth, colour-coded proton+deuteron, and all particles) are produced for Sample I and Sample II in both the full downstream-sum and the data-matched four-layer definitions. Penetration-depth distributions (L_{stop}: deepest physical layer with a deposit above a stated threshold, with a threshold-stability scan) are produced per species and per sample, together with the fraction of events reaching each downstream readout position (B4/B6/B8) and the beam-data Sample I versus Sample II stopping-layer overlay from the producer's threshold-pass logic. Beam-data axes remain amplitude proxies (not calibrated energy); B2 saturation is shown only as the analysis-contract flag, with no numeric hardware level asserted.

The MC panels are Geant4 deposited-energy truth views from a 2,000,000-event scale-up of the regenerated campaign (`geant4/manifests/cmc_2m_regenerated_20260814.json`; identical sampler, geometry and runtime stack as the authorising 100,000-event campaign). Because both campaigns run the default engine seed over the same sequential stream, the scale-up replicates the authorising campaign as a verified sequential prefix: the leading B-stack-entrance event rows are identical event-by-event, and the added statistics only densify the same truth topology. The set is descriptive truth topology, not a detector-response prediction: readout thresholds, light collection and digitisation are not in the loop, so no numeric agreement with the amplitude-domain data figures is claimed or implied. Beam-data figures remain GATED under the same Cycle-3 gates as Figure 7/8; the MC truth views carry SIMULATION_RESULT status and promote no quarantined number. Artifact bundles: `reports/paper_618_species_penetration_2m_20260814T1449Z/` (final set) and `reports/paper_618_species_penetration_20260814T1615Z/` (authorising 100k thin-sample record).

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/final/618_mc_truth_full_I.pdf`

Species-resolved truth topology from the 2M regenerated campaign; descriptive MC only, not a detector-response prediction.

Figure 8: MC truth ΔE–E, Sample I, full downstream sum; proton/deuteron/combined/all panels (#618, SIMULATION_RESULT).

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/final/618_mc_truth_full_II.pdf`

Same provenance as the Sample I figure; unweighted event counts shown on panels.

Figure 9: MC truth ΔE–E, Sample II, full downstream sum; same panel structure (#618, SIMULATION_RESULT).

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/final/618_mc_truth_4layer_I.pdf`

Uses only the same physical readout positions as the data (B2/B4/B6/B8).

Figure 10: MC truth ΔE–E, Sample I, data-matched four-layer definition (#618, SIMULATION_RESULT).

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/final/618_mc_truth_4layer_II.pdf`

Same definition as the Sample I data-matched view.

Figure 11: MC truth ΔE–E, Sample II, data-matched four-layer definition (#618, SIMULATION_RESULT).

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/final/618_penetration_I.pdf`

Threshold-stability scan and reaching fractions recorded in the artifact bundle.

Figure 12: MC penetration depth L_{stop}, Sample I: proton, deuteron, shape-normalised overlay, all particles (#618, SIMULATION_RESULT).

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/final/618_penetration_II.pdf`

Deepest-layer threshold and its stability variants documented in the bundle tables.

Figure 13: MC penetration depth L_{stop}, Sample II, same panel structure (#618, SIMULATION_RESULT).

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/gated/618_data_amplitude_I_II.png`

Amplitude proxies, not calibrated energy; B2 saturation shown as the analysis-contract flag only; Cycle-3 gates apply as for Figure 7/8.

Figure 14: Beam-data amplitude analogue ΔE–E, Samples I and II on identical axes (#618, GATED).

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/gated/618_data_penetration_I_II.png`

Uses the producer's threshold-pass logic; no new ADC cut is introduced here.

Figure 15: Beam-data stopping-layer overlay, Sample I versus II, amplitude domain (#618, GATED).

8 Monte Carlo prediction of light collection and transport

8.1 Stage-resolved response

The optical model should report separately the production of scintillation photons, WLS absorption, WLS re-emission, guided/transported photons, sensor arrivals, primary SiPM avalanches, correlated avalanches, charge and ADC response. Useful conditional efficiencies include

$$\epsilon_{\mathrm{WLS\,capture}} = \frac{N_{\gamma,\mathrm{WLS\,abs}}}{N_{\gamma,\mathrm{scint}}}, \quad \epsilon_{\mathrm{transport}} = \frac{N_{\gamma,\mathrm{sensor}}}{N_{\gamma,\mathrm{WLS\,emit}}}, \quad \epsilon_{\mathrm{PDE}} = \frac{N_{\mathrm{primary\,aval}}}{N_{\gamma,\mathrm{sensor}}}.$$

These ratios should be studied versus hit position, species/deposit regime and model nuisance parameters. Their denominators must be named explicitly: an efficiency conditional on WLS emission is not an absolute scintillator-to-sensor efficiency.

8.2 Historical July calibration grid

The July single-stave grid contains five proton/deuteron points and historically produced an order-ten detected-PE per visible-MeV scale. It is useful as a regression/history object, but it is not the nominal publication model.

> **PUBLICATION HOLD.** The exact July calibration revision predates multiple source-level fixes. In particular, inner and outer fibre claddings both used the shared NIST `G4_PLEXIGLASS` material and attached different refractive-index tables, allowing the later assignment to overwrite the intended inner-clad optical properties. The campaign also predates explicit WLS fluorescence-yield/multiplicity, separated scintillator/Y-11 attenuation controls, corrected reflector-surface semantics, source-bound fibre-to-SiPM interface modelling and a single canonical SiPM response state. Its historical "PE/MeV" denominator is the Birks-visible variable, not unquenched deposited energy. Issue #1303 therefore requires regeneration before any numerical light-yield result is promoted.

8.3 Regenerated 2026-08 calibration grid (current model)

Issue #1303 regenerated the five-point grid (p60/p100/p140/d70/d110, 2000 events per point, seed-bound, distinct inner/outer cladding materials, SHA-256-bound inputs and manifests). The pooled Birks-visible light yield is 11.78 PE/MeV$_{\mathrm{vis}}$ (offset 0.65 PE, $r^2 = 0.976$, $n = 10{,}000$); per-point E_{vis} yields are flat within their bootstrap intervals (11.78–11.84), while E_{raw} yields (8.6–10.6) split by species through the Birks quenching ratio ($E_{\mathrm{vis}}/E_{\mathrm{raw}} \approx 0.81$ for 110 MeV deuterons). Stage accounting gives mean capture $\epsilon_{\mathrm{WLS}} \approx 0.19$, transport ≈ 0.021, detection-given-arrival ≈ 0.30, and a saturated-PE-to-detected ratio ≈ 0.96 at the readout channel. These numbers are MC_MODEL_DEPENDENT: the optical/SiPM nuisance envelope (PAPER-A07/A08) is not yet propagated, so the July grid remains the historical object and no beam-data calibration is claimed. Machine-readable source tables accompany every figure (`publication/tables/final/`).

The historical values and plots remain available under `publication/figures/gated/` and the source report for correction history only.

8.4 Required nominal optical campaign

A publication campaign must bind a clean source revision and build receipt; preserve E_{raw} and E_{vis} separately; record stage counters; use the declared one-fibre/one-end design channel as the nominal response only after the installed hardware configuration is source-bound under issue #1296; retain alternative fibre/end observables as model controls where useful; scan or state the support in longitudinal/transverse position and angle; and propagate independent high-impact optical nuisance families rather than re-tuning the total mean response back to a historical target.

9 Energy reconstruction and resolution

9.1 Measurand definition

Energy resolution is not a single detector-independent number. For the single-stave response chain, two physically distinct targets are available:

$$E_{\mathrm{raw}} \equiv \text{unquenched Geant4 energy deposited in the scintillator,} \tag{1}$$

$$E_{\mathrm{vis}} \equiv \text{Birks-visible/quenched energy used to generate scintillation light.} \tag{2}$$

A full range-stack reconstruction may later target incident charged-particle kinetic energy from amplitudes and stopping depth. These estimands must not share one bare E_{dep} label.

For a chosen target E_T, define

$$r = \frac{E_{\mathrm{reco}} - E_T}{E_T}.$$

A publication result should report median bias, σ_{68}, RMS, tail fraction and uncertainty intervals on held-out populations, as functions of target energy and relevant phase-space variables. The uncertainty unit must reflect the independent simulation/run conditions used to establish transfer, not merely the number of events generated under a few shared grid settings.

9.2 Status of the current held-out result

PUBLICATION HOLD. The 8.9% held-out σ_{68} result remains quarantined. Energy semantics are now enforced API-wide (#1338): E_{raw} (unquenched Geant4 scintillator deposit) and E_{vis} (Birks-visible quenched energy) are distinct estimands. The held-out producer was rewritten (#1339) with proper estimand definition and propagated uncertainty, and the underlying optical grid was regenerated (#1303, complete 2026-08). On the regenerated five-point grid with the E_{vis} target: held-out median bias +0.48% [+0.29, +0.62], $\sigma_{68} = 6.26\%$ [6.16, 6.35], tail fraction 0.2%; the E_{raw} estimand (deliberately mis-specified pooled-linear response) gives bias +10.4%, $\sigma_{68} = 9.0\%$, tail 16.9% and serves as the negative control, alongside the shuffled-target control reported in the result manifest. Bootstrap 16–84% intervals use 500 resamples of the held-out population. The result remains MC_MODEL_DEPENDENT and quarantined from VALIDATED promotion: the optical/SiPM nuisance envelope is not yet propagated, held-out coverage spans two operating points only, and impact-position transfer is scoped to the central line ($y = 0$, #1092). No promotion to VALIDATED is authorized.

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/gated/edep_reconstruction_heldout.png`

GATED: wrong/ambiguous energy semantics, incomplete uncertainty and superseded optical campaign.

Figure 16: Quarantined Figure 11 from the historical held-out reconstruction study.

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/gated/edep_reconstruction_heldout_E_vis.pdf`

GATED: MC_MODEL_DEPENDENT (optical/SiPM nuisance envelope not propagated; two held-out operating points; simulation, not beam data).

Figure 17: Held-out energy reconstruction on the regenerated #1303 optical grid (E_{vis} estimand): residual distribution for held-out deuteron_110 and proton_60 with bootstrap 16–84% intervals; median bias +0.48%, $\sigma_{68} = 6.26\%$, tail fraction 0.2%.

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/gated/edep_reconstruction_heldout_E_raw.pdf`

GATED: deliberately mis-specified response control; shown to demonstrate estimand separation, not as a performance claim.

Figure 18: Negative control: the same reconstruction evaluated against the E_{raw} (unquenched) target with the pooled E_{vis}-calibrated linear response (median bias +10.4%, $\sigma_{68} = 9.0\%$, tail fraction 16.9%).

9.3 Required publication analysis

The replacement study should preregister and freeze training/calibration, model-selection and final-evaluation populations before fitting; include multiple well-separated target-energy regimes; validate particle-species transfer without truth-label tuning; test longitudinal/transverse position and angle support; compare transparent linear/monotonic baselines before any ML model; and propagate optical/SiPM model nuisance families separately from finite-simulation statistics. Bootstrap or hierarchical intervals should resample the independent grid/run unit appropriate to the claim. A shuffled-target or deliberately mis-specified response control should fail the reconstruction test, and occupancy/saturation diagnostics should be checked on known-answer inputs with a signed definition. If a resolution parametrisation is fitted, the available independent energy grid must identify its terms.

A full-stack incident-energy estimator may use the amplitude vector $\mathbf{A} = (A_{B2}, A_{B4}, A_{B6}, A_{B8})$ and a source-bound penetration/range observable, but it remains downstream of material-budget, trigger/selection and per-channel response closure.

10 Discussion

Existing selected-data studies indicate that the declared Sample-I and Sample-II populations occupy the B stack differently in depth. The difference has now been carried through the pre-threshold 8-by-16 event reconstruction with run-block uncertainty and a baseline/threshold scan, providing the paper's most direct experimental range-stack observation: its direction is stable across the 0–1000 ADC scan while its magnitude is strongly threshold-compressed. Even so, the first interpretation remains a run-family dependence: a unique trigger-causation statement additionally requires the primary hardware trigger and run log to be source-bound.

Sparse longitudinal sampling is a central detector-design issue. An alternating four-position readout can preserve a coarse penetration pattern while losing the local sequence of energy loss near the Bragg rise. This creates a concrete aliasing mechanism: changes in particle energy, incidence angle or passive material can move the high-dE/dx region between instrumented positions. A quantitative PID or range estimator based on sparse ΔE–E information therefore requires either sufficiently dense longitudinal instrumentation or an independently validated model of the missing-layer response, with the readout phase treated as a nuisance rather than as hidden truth.

The timing analysis gives a complementary negative result. The located waveform schema supports an approximately 38-ns B4–B6 pair-residual diagnostic in the current reconstructed subset, but it does not support an intrinsic sub-nanosecond stave-resolution measurement. Component identity, timing reference, run dependence and channel covariance must be established before any deconvolution to a single-stave resolution is attempted. This limitation is scientifically useful because it states what a future campaign must record or calibrate: a source-bound timebase, sufficient waveform window, stable component assignment and an independent timing reference or covariance model.

The optical-model audit shows why a detector-response study must distinguish a causal response graph from a tuned total efficiency. WLS fluorescence multiplicity, trapping and attenuation, fibre-end coupling, SiPM photon-detection response, microcell occupancy/recovery, correlated noise and electronics can compensate one another in an integrated charge or amplitude. Stage-resolved counters, phase-space scans and independent bench/hardware constraints are therefore more informative than tuning a single PE/ADC conversion to one dataset.

The present limitations are coupled rather than independent. Geometry and run/trigger uncertainty affect which particles enter the selected data population; source-model and event-measure uncertainty affect the MC comparison; sparse readout affects the observability of a local stopping-power rise; and optical/electronics assumptions affect any conversion from deposited energy to charge or ADC. The final paper should therefore report the direct data observables first and introduce model-dependent detector-performance claims only after these nuisance layers have been propagated through the same final population definitions.

11 Conclusions

The CCB test-beam analysis provides a controlled test of a segmented plastic-scintillator range stack for low-energy charged particles. The regenerated pre-threshold 8-by-16 event population (33 runs, 1.1M events, run-block bootstrap) confirms different longitudinal B-stack populations for the declared Sample-I and Sample-II run families — entrance concentration 87.4% versus 72.7% in B2 and a deepest-stave share 7.6× larger in Sample II — but the quantitative depth-profile claim remains gated: the profile shape is strongly threshold-dependent and the primary hardware-trigger record is not source-bound. The analysed B readout samples only alternating longitudinal positions, so any range or energy-loss estimator is sensitive to where the stopping-power rise falls relative to instrumented positions. The intended telescope-style amplitude observable is $\Delta E = A(B2)$ and $E = A(B4) + A(B6) + A(B8)$; a B2–B4 plot is a two-channel diagnostic rather than the complete downstream-energy proxy.

The located 8-by-16 waveform product supports a negative timing conclusion rather than a detector-resolution number. In the component-safe rerun, the B4–B6 pair residual has a format-limited central width of 8.7 ns (superseding an earlier 38 ns direct-mode diagnostic) and is affected by the finite waveform representation, unvalidated multi-component identity and unresolved timing-reference/covariance terms. Historical sub-nanosecond simulation or incompatible-waveform results must therefore not be transferred to this beam-data product.

A single-stave Geant4 optical model remains valuable as a response framework. The historical July calibration campaign is superseded as the nominal model; the grid has been regenerated on the repaired source with explicit E_{raw} and E_{vis} semantics, and the held-out reconstruction has been rebuilt on it with propagated uncertainty. These regenerated results remain MC_MODEL_DEPENDENT and quarantined from validated promotion: the optical/SiPM nuisance envelope is not yet propagated, held-out coverage spans two operating points only, and impact-position transfer is scoped to the central line, so they do not authorize absolute light-yield or resolution claims.

The paper is therefore not yet a quantitative detector-performance publication. The shortest defensible path to submission is to close the source-bound hardware/run/polarity and 8-by-16 data contracts first, produce the final longitudinal beam-data result, establish a provenance-complete MC event measure, close the remaining ΔE–E gates (hardware source-binding of the channel-to-layer map and of the saturation/transfer threshold) and deliver the final figure package, and then decide whether the optical/energy programme is mature enough to remain in the same paper. If it is retained, the remaining optical steps are propagation of response-stage nuisance families and a broader genuinely held-out reconstruction study across more operating points and impact positions; the regenerated grid already separates E_{raw} and E_{vis}. Every headline result should then be bound to its machine-readable source table, uncertainty, canonical claim status and exact reviewed source head before submission.

12 Reproducibility and figure-generation requirements

Every quantitative paper figure must be generated from a tracked machine-readable result or source table. The producing record must include source commit, producer hash, input paths and SHA-256 values, event/run population, selection, weights, uncertainty method and evidence class. No headline number should be manually copied into a plotting script or caption.

Data and MC must use compatible observable definitions. Beam ADC amplitudes and truth-level Geant4 energy deposits may be shown in separate panels, but they must not be placed on a shared calibrated energy axis without a validated response transformation. Weighted MC results must state the weight semantics and report $\sum w$, $\sum w^2$, effective sample size and weight pathologies relevant to the estimand.

Timing and energy calibrations must be fit on declared calibration/training populations and evaluated on held-out runs or simulation populations without retuning. Resampling must preserve the physical/statistical source unit, normally DAQ run/event for beam data and generator event for MC, rather than assuming every exported hit/pulse row is independent.

The canonical publication workflow is

$$\begin{aligned}\text{physics/provenance contract} &\rightarrow \text{result + manifest + source table}\\ &\rightarrow \text{claim ledger} \rightarrow \text{figure/table}\\ &\rightarrow \text{manuscript} \rightarrow \text{adversarial review.}\end{aligned}$$

An issue is not publication-complete when code alone runs successfully.

A Publication status and open gates

This appendix is part of the working PDF so that exported copies cannot lose the scientific readiness state. It is collaboration-review metadata rather than a substitute for the final scientific results narrative; before journal submission, detailed issue bookkeeping should move to supplementary reproducibility material.

Gate	Publication consequence
#1296	Hardware/BOM/run record: source-bind installed stave, channel map, target, trigger and survey/run settings.
#962 / #1045	Run-role and trigger response: prevents over-interpreting Sample-I/Sample-II differences as pure trigger effects.
#869 / #954	Readout-map and waveform polarity: required before authorising event-level amplitude observables.
#956 / #1321	Regenerate beam/MC ΔE–E from the authorising pre-threshold 8×16 event product and corrected MC layer namespaces.
#1053 / #1179 / #1311	MC event measure, source uncertainty and production provenance. The deterministic direct-CDF sampler defect tracked by #1178 is repaired/closed; that bounded software closure does not authorise the historical paper MC.
#1088 / #1303 / #1322	WLS photophysics and regeneration of the single-stave optical grid after superseded source defects, followed by replacement optical/energy figures.
#1302 / #1297	Raw deposited energy vs Birks-visible energy semantics and a corrected held-out reconstruction study.
#1317–#1320	Final setup, beam-data longitudinal-profile, MC longitudinal-profile and timing figure packages remain pending.
#1304 / #1299	Single scientific source of truth for claim status across figures, WIKI and manuscript, plus stale-document reconciliation.
#1301 / #1305	Umbrella submission and merged-state fail-closed gates.

As of the 2026-08-14 current-head audit, the authorising final-figure, final-figure-source-data and final-result-table namespaces contain no non-README scientific artifacts. The build validator checks the controlled package structure and hold state; it is not yet a numerical/provenance submission-readiness validator.

The Cycle-3 audit documents the detailed failure modes under `chatgpt_todo/PAPER_SUBMISSION_AUDIT_CYCLE3_20260812.md` and its addenda. The current-head dependency order and section-by-section review are recorded under `chatgpt_todo/PAPER_COMPLETION_AUDIT_20260814.md`.

B Primary repository evidence surfaces

The working paper is derived from, and must remain synchronized with, the following repository evidence surfaces:

- `docs/claim_ledger.csv` – canonical scientific claim status;
- `paper/hardware_bom.csv` – publication hardware truth/status surface;
- `paper/figures.yaml` – historical figure registry, currently subject to governance repair under #1304;
- `chatgpt_todo/PAPER_SUBMISSION_AUDIT_CYCLE3_20260812.md` and addenda – current fail-closed review;
- `reports/studies/data_side/REPORT.md` – located raw beam-data audit and timing limitation;

- reports/1781181864.166832.35d806b2__s21_geant4_source_review/REPORT.md – source-level CCB Geant4 review;
- docs/adr/ADR-WLS-FLUORESCENCE-YIELD-UNVERIFIED.md – absolute WLS-yield boundary;
- docs/adr/ADR-SIPM-PHYSICS-BLOCKED-WAVEA-LANE01.md – SiPM-response boundary.

The previous Markdown working manuscript is retained under publication/source/manuscript_working_20260812.md for provenance. The chapterised TeX files in this folder are the canonical editing surface going forward.

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